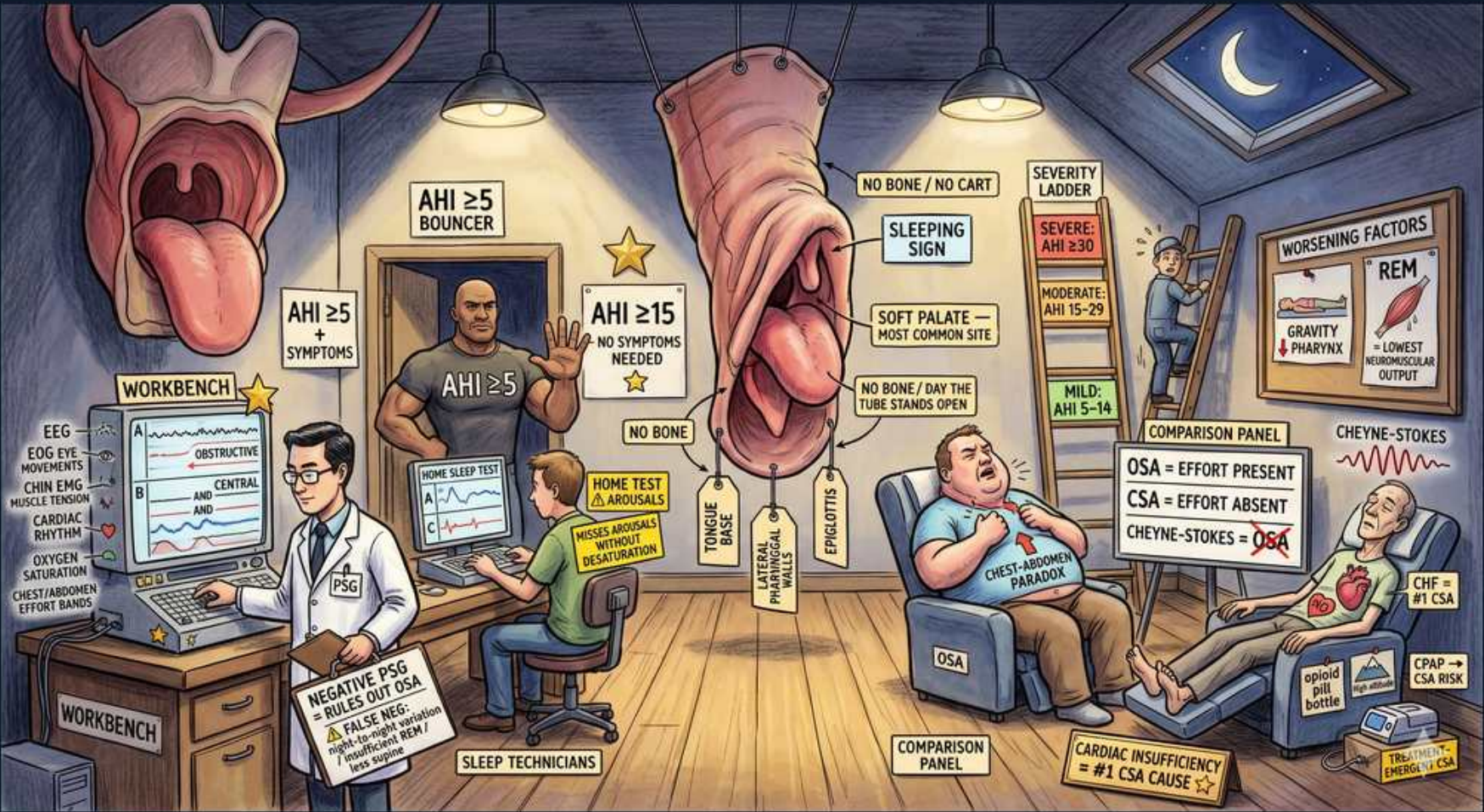


Sleep Apnea — Diagnosis & Severity (AHI)



1. Polysomnography workbench

WHAT YOU SEE

A lab 'WORKBENCH' desk marked with a GOLD STAR, its monitor scrolling EEG, oxygen-saturation, and chest/abdomen effort-band tracings — the gold star flags in-lab PSG as the gold-standard test.

WHAT IT ENCODES

In-lab attended polysomnography (PSG) is the GOLD-STANDARD diagnostic test for obstructive sleep apnea. It simultaneously records EEG (sleep staging/arousals), EOG (eye movements), chin/leg EMG (tone, periodic limb movements), ECG, airflow, thoracoabdominal effort belts, and pulse oximetry. By capturing respiratory EFFORT, PSG distinguishes obstructive events (effort present against a closed airway) from central events (no effort) — the single most important diagnostic distinction.

BOARD TRAP

WRONG answer: ordering a daytime ABG or spirometry to 'diagnose OSA.' Those assess gas exchange and obstructive lung disease, not sleep-disordered breathing. The discriminator is that OSA is defined by events during SLEEP, quantified only by a sleep study (PSG or home test) — pulmonary function testing cannot diagnose it.

2. Sleep technician

WHAT YOU SEE

The white-coated figure standing at the workbench holding a clipboard stamped 'PSG' — the technician scoring the overnight study.

WHAT IT ENCODES

The technician scores the AHI = (Apneas + Hypopneas) per hour of sleep. An APNEA is $\geq 90\%$ airflow drop for ≥ 10 seconds; a HYPOPNEA is a $\geq 30\%$ airflow drop for ≥ 10 seconds accompanied by $\geq 3\%$ (or $\geq 4\%$) desaturation or an arousal. The RDI additionally counts respiratory-effort-related arousals (RERAs). Attended scoring is why in-lab PSG outperforms unattended home testing for subtle, arousal-based disease.

BOARD TRAP

WRONG answer: assuming any breathing pause counts. An apnea must last ≥ 10 seconds and a hypopnea requires the desaturation/arousal criterion — brief normal pauses do not qualify. The discriminator is that AHI is per HOUR OF SLEEP (not per hour in bed), which is why total sleep time from EEG matters and home tests (which estimate from recording time) can underestimate.

3. AHI ≥ 5 + symptoms

WHAT YOU SEE

A hanging placard reading 'AHI ≥ 5 + SYMPTOMS' — the bouncer only lets you in at 5 if you also show the 'ID' of symptoms.

WHAT IT ENCODES

OSA is diagnosed when AHI ≥ 5 /hr WITH symptoms or comorbidities — excessive daytime sleepiness, loud snoring, witnessed apneas/gasping, nonrestorative sleep, fatigue, or associated conditions (hypertension, mood disorder, type 2 diabetes, AF, stroke). At AHI 5–14 the symptom requirement is mandatory; asymptomatic AHI 5–14 alone does not meet diagnostic criteria.

BOARD TRAP

WRONG answer: diagnosing OSA in an asymptomatic patient with AHI of 8. Below 15, symptoms (or qualifying comorbidity) are REQUIRED. The discriminator: AHI 5–14 needs symptoms; AHI ≥ 15 stands alone regardless of symptoms.

4. AHI ≥ 5 bouncer

WHAT YOU SEE

A muscular bald nightclub bouncer, hand raised to stop you, his chest labeled 'AHI ≥ 5 ' and topped with a GOLD STAR — AHI 5 is the entry threshold guarding the diagnosis.

WHAT IT ENCODES

AHI 5 is the entry threshold ('bouncer') — below 5 events/hr is normal and excludes OSA. To get past the bouncer at 5–14, the patient must show their 'ID' (symptoms/comorbidity). The bouncer concept also flags that pre-test screening (STOP-BANG, Epworth) raises suspicion but only the AHI grants entry to the diagnosis.

BOARD TRAP

WRONG answer: ruling OSA out because AHI is exactly 5 — that meets the threshold, not below it. Conversely, an AHI of 3 is normal and should not be called 'mild OSA.' The discriminator is the hard cutoff: < 5 normal, ≥ 5 (with symptoms) qualifies.

5. AHI ≥ 15 — no symptoms needed

WHAT YOU SEE

A placard reading 'AHI ≥ 15 — NO SYMPTOMS NEEDED' wearing its own GOLD STAR — at 15 the number alone earns the diagnosis, no symptom 'ID' required.

WHAT IT ENCODES

AHI ≥ 15 /hr diagnoses moderate-to-severe OSA EVEN IN AN ASYMPTOMATIC patient, because the cardiovascular and metabolic risk at this burden warrants the diagnosis and treatment regardless of reported sleepiness. This is a favorite board distinction — the ≥ 15 threshold removes the symptom requirement.

BOARD TRAP

WRONG answer: withholding the OSA diagnosis (or CPAP) because an AHI-22 patient 'feels fine' and denies sleepiness. AHI ≥ 15 is diagnostic on its own. The discriminator: at ≥ 15 symptoms are irrelevant to the diagnosis; below 15 they are mandatory.

6. Severity ladder

WHAT YOU SEE

The 'SEVERITY LADDER' with rungs

WHAT IT ENCODES

Severity is graded by AHI: MILD 5–14, MODERATE 15–29, SEVERE ≥ 30 events/hr. Severity drives management intensity: moderate-to-severe disease (AHI ≥ 15 , or ≥ 5 with significant symptoms/comorbidity) is the clearest indication for CPAP, while mild disease may be managed with weight loss, positional therapy, or an oral appliance.

BOARD TRAP

WRONG answer: mixing up the cut points (e.g., calling AHI 20 'mild'). Memorize 5 / 15 / 30. The discriminator: the SAME number 15 is both the moderate-severity floor AND the symptom-independent diagnostic threshold — a deliberately tested overlap.

7. Mild 5–14

WHAT YOU SEE

'MILD: 5–14' lowest rung

WHAT IT ENCODES

Mild OSA = AHI 5–14 events/hr, and (because it is <15) requires symptoms or comorbidity to be diagnosed at all. Mild, obesity-related disease is the group most likely to respond to conservative measures — weight loss, avoiding alcohol/sedatives, positional therapy, and mandibular advancement devices — though CPAP remains an option if symptomatic.

BOARD TRAP

WRONG answer: reflexively prescribing CPAP for every mild case. For mild OSA, weight loss/oral appliance/positional therapy are reasonable first steps, especially if the patient declines CPAP. The discriminator: severity (and symptoms) tailor therapy — CPAP is most strongly indicated for moderate-severe disease.

8. Moderate 15–29

WHAT YOU SEE

'MODERATE: 15–29' middle rung

WHAT IT ENCODES

Moderate OSA = AHI 15–29 events/hr. Being ≥ 15 , it is diagnostic without symptoms and is a firm indication for CPAP, which improves sleepiness, blood pressure, and quality of life. Moderate disease is the typical threshold above which untreated OSA meaningfully raises cardiovascular and motor-vehicle-accident risk.

BOARD TRAP

WRONG answer: requiring symptoms before diagnosing or treating an AHI-18 patient. At ≥ 15 the disease is established regardless of symptoms, and CPAP is indicated. The discriminator is the 15 floor separating mild (symptom-dependent) from moderate (symptom-independent).

9. Severe ≥ 30

WHAT YOU SEE

'SEVERE: AHI ≥ 30 ' top rung

WHAT IT ENCODES

Severe OSA = AHI ≥ 30 events/hr — the strongest indication for CPAP and the group with the highest risk of resistant hypertension, atrial fibrillation, stroke, pulmonary hypertension/cor pulmonale, and accidents. Severe disease should be treated even if the patient minimizes symptoms, and adherence should be aggressively supported.

BOARD TRAP

WRONG answer: offering an oral appliance or UPPP as first-line for severe OSA. Oral appliances and surgery are inferior for severe disease; CPAP is first-line. The discriminator: at AHI ≥ 30 , CPAP (not a dental device or surgery) is the standard-of-care initial therapy.

10. Home sleep test

WHAT YOU SEE

A younger seated technician at a laptop tagged 'HOME SLEEP TEST', beside a red warning sign 'HOME TEST $\square$ MISSES AROUSALS WITHOUT DESATURATION' — the home rig has no EEG so arousal-only events slip past it.

WHAT IT ENCODES

Home sleep apnea testing (HSAT) is an acceptable alternative to in-lab PSG ONLY in patients with HIGH pre-test probability of moderate-to-severe UNCOMPLICATED OSA (no significant cardiopulmonary disease, neuromuscular disease, hypoventilation, or suspected central apnea). HSAT measures airflow, effort, and oximetry but no EEG, so it cannot detect arousals or stage sleep, and it estimates AHI over recording time rather than true sleep time — biasing the AHI downward.

BOARD TRAP

WRONG answer: using a home test in a patient with heart failure, suspected central sleep apnea, COPD, or hypoventilation. HSAT is validated only for uncomplicated high-probability OSA; complex patients need in-lab PSG. The discriminator is the comorbidity profile — comorbid cardiopulmonary/neuromuscular disease mandates full PSG.

11. Negative PSG — repeat

WHAT YOU SEE

The PSG doctor holding a banner 'NEGATIVE PSG $\rightarrow$ RULES OUT OSA' stamped 'FALSE NEG' in caution — a normal home study in a high-suspicion patient is a false-negative trap, not a true rule-out.

WHAT IT ENCODES

Because HSAT systematically underestimates AHI (no EEG arousals, recording-time denominator, night-to-night variability), a NEGATIVE home study in a patient with high clinical suspicion does NOT rule out OSA and must be followed by in-lab attended PSG. A single negative PSG in a very symptomatic patient may also warrant repeat testing.

BOARD TRAP

WRONG answer: concluding 'OSA excluded' after a normal home study in a classic high-suspicion patient (obese, snoring, witnessed apneas, sleepy). Negative HSAT in high pre-test probability requires escalation to full PSG. The discriminator: a negative home test is non-diagnostic when suspicion is high — it only confirms OSA, it cannot exclude it.

12. Worsening factors: REM & supine

WHAT YOU SEE

'WORSENING FACTORS — GRAVITY (supine), REM' panel

WHAT IT ENCODES

Obstructive events are worst in REM sleep and in the SUPINE position. REM sleep produces near-complete skeletal muscle atonia (including the genioglossus/pharyngeal dilators), so airway collapse and the deepest, longest desaturations cluster in REM. Supine posture lets gravity drop the tongue and soft palate posteriorly. This is the rationale for POSITIONAL therapy in patients with predominantly supine or positional OSA.

BOARD TRAP

WRONG answer: dismissing a study as negative when overall AHI is borderline but the supine/REM AHI is markedly elevated. Position- and stage-dependent OSA is real disease and may be missed if the patient slept little supine or in REM. The discriminator: REM atonia and supine gravity explain why event timing matters and why positional therapy can help.